

GROUP B OFFICERS ACADEMY (WWW.GROUPBOFFICERSACADEMY.COM)

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WAGON TURN ROUND (WTR)

1. Introduction

Wagon Turn Round is one of the most important performance indices used by Indian Railways to measure the **efficiency of wagon utilisation**.

It directly reflects how quickly a wagon completes a full loading–unloading–return cycle and becomes available again for reuse.

2. Definition

Wagon Turn Round (WTR)

→ “The interval between two successive loadings, calculated from the time a wagon is placed for loading until it again becomes available for reloading.”

This represents the **total time a wagon spends in one complete operational cycle**.

3. Why Individual Calculation is Not Practicable

- Indian Railways handles **lakhs of wagons** simultaneously.
- Wagons are in constant movement across several divisions.
- Tracking individual wagon time cycles is operationally impossible.
- Hence a **statistical formula** is used for system-wide estimation.

4. Statistical Formula for Wagon Turn Round

$$\text{Wagon Turn Round (T)} = \frac{S}{L + R}$$

Where:

Symbol Meaning

- S** Effective daily wagon holding (midnight balance)
- L** Total number of wagons loaded (Division + Transshipment points)

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Symbol Meaning

R Total number of inward loaded wagons received from other Divisions/Railways

5. What is NOT Included in 'S' (Wagon Holding)?

When calculating **effective wagon holding**, the following wagons are excluded:

1. **Sick wagons** (under repair)
2. **Wagons in POH or waiting for shops**
3. **Departmental wagons**
4. **Wagons lent for departmental use**
5. **Wagons used for coaching stock purposes**

Only **traffic-serviceable wagons** are counted.

6. Interpretation of L and R

L — Total Wagons Loaded

Includes:

- All wagons loaded on the Division
- Wagons loaded at Transshipment Points (e.g., MG to BG transshipment loads)

R — Loaded Wagons Received

Loaded wagons arriving from:

- Other Divisions
- Other Railways

Thus **(L + R)** represents the **total number of loaded wagons handled** in a day.

7. Example:

Given:

- Wagons loaded in Division = **350**

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- Out of 350, loaded through Transshipment = **50**
- Inward loaded wagons received = **150**
- Effective wagon holding (midnight balance) = **2250**

Calculation:

$$T = \frac{S}{L + R} = \frac{2250}{350 + 150} T = \frac{2250}{500} = 4.5 \text{ days}$$

WTR = 4.5 days

9. Factors Affecting Wagon Turn Round

(Very Important for AOM Viva & MCQs)

A. Loading/Unloading Efficiency

- Terminal handling time
- Availability of labour & equipment
- Mechanised loading/unloading

B. Yard & Siding Operations

- Shunting delays
- Placement and withdrawal time
- Inter-departmental coordination

C. Line Capacity Constraints

- Congestion on main routes
- Speed restrictions
- Crew availability

D. Rolling Stock Factors

- Sickness rate
- Maintenance availability

E. Administrative Factors

- Planning of empties
- Monitoring by control

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- Priority to high throughput commodities

10. Measures to Reduce Wagon Turn Round

(High-value points for exams)

1. **Increase terminal handling capacity**
(mechanisation, additional lines)
 2. **Efficient yard management**
(minimise shunting, reduce idle time)
 3. **Better planning of empties**
(real-time allocation by control)
 4. **Improve line capacity**
(remove bottlenecks, better scheduling)
 5. **Reduce sick wagons**
(preventive maintenance)
 6. **Quick release of wagons after unloading**
 7. **Use of IT systems**
(FOIS for real-time monitoring)
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